Rojae Mighty
Stony Brook, NY · rojae.mighty@stonybrook.edu
rojaemighty.com · linkedin.com/in/rojaemight

July 15, 2026

Hiring Manager
Tower Research Capital
New York, NY

Re: Quantitative Developer Intern

Dear Hiring Manager,

I am excited to apply for the Quantitative Developer Intern position at Tower Research Capital. I am an undergraduate physics student at Stony Brook University with experience developing computational analyses, working with complex datasets, and building simulation-based research workflows. My background in C++, Python, Unix/Linux, and quantitative modeling has prepared me to contribute to a technically demanding environment where reliable software and careful analysis matter.

Through my research at Brookhaven National Laboratory and the Center for Frontiers in Nuclear Science, I use Monte Carlo event generation, ROOT, C++, and Python to study electron–ion collisions and nuclear effects. My work includes implementing a spatially dependent parton distribution function, generating and analyzing pseudodata for vector-meson production with the SARTRE Monte Carlo framework, and evaluating how model assumptions affect physical observables. These projects have strengthened my object-oriented programming, debugging, statistical analysis, and problem-solving skills while teaching me to validate computational results carefully.

I am also developing a market microstructure forecasting project using limit order book data. The project involves building a Python pipeline, engineering features such as order-flow imbalance and spread dynamics, forecasting short-horizon mid-price returns and volatility, and evaluating models with walk-forward validation and transaction-cost-aware backtesting. This work has given me a practical introduction to financial data and the challenges of building robust systems for noisy, time-dependent environments.

Tower's focus on research, technology, and fast, data-driven decision-making strongly appeals to me. I would be eager to apply my experience with scientific software and quantitative modeling to financial markets, learn from Tower's developers and researchers, and contribute code that is efficient, testable, and useful in production-oriented research. I would bring curiosity, attention to detail, and a disciplined approach to solving unfamiliar technical problems.

Thank you for considering my application. I would welcome the opportunity to discuss how my computational research experience, programming background, and growing interest in quantitative finance could contribute to Tower Research Capital.

Sincerely,

Rojae Mighty